AFDB · DATA INNOVATION FOR AFRICA (DI4A) · HANDS-ON GUIDE

Generate a GitHub API Key

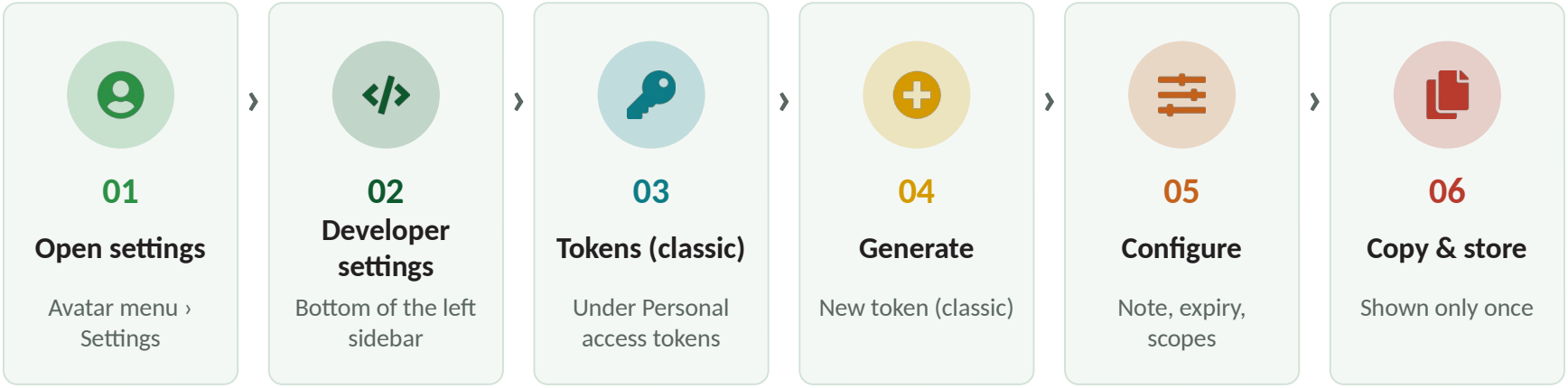
Create a personal access token (classic) in six steps



AFRICAN DEVELOPMENT BANK
Data Innovation
for Africa (DI4A)

African Development Bank (AfDB)
Data Innovation for Africa (DI4A)

From your profile to a working token



Treat the token like a password. A classic token can be used instead of your password for Git over HTTPS and to authenticate to the GitHub API.

Where GitHub fits in your data work

Data infrastructure

Data access and governance; Cloud, GitHub and Colab working together.

Team projects

Shared repositories for code, data and project deliverables.

Hands-on notebooks

Machine learning for GDP nowcasting; trade nowcasting with PortWatch.

Sharing results

Version, review and publish your team's code and results.

The token authenticates your scripts and notebooks to the GitHub API.

BEFORE YOU START · NO GITHUB ACCOUNT YET?

Create a GitHub account with your Gmail address

About 5 minutes. Keep your Gmail inbox open: GitHub sends you a code to confirm the address.



Two ways to sign up with Gmail

A

FASTEST WITH GMAIL

Continue with Google

Use your Gmail account to sign in to GitHub. No separate GitHub password to create.

1. Open github.com/signup
2. Click “Continue with Google”
3. Pick your Gmail account, then follow the prompts

B

Sign up with your Gmail address

Create a classic GitHub login: your Gmail address plus a new password.

1. Open github.com/signup
2. Enter Gmail address, password, username
3. Confirm with the code sent to Gmail

Either way, use an address you check regularly: without a verified email you cannot create repositories.

Fill in the sign-up form

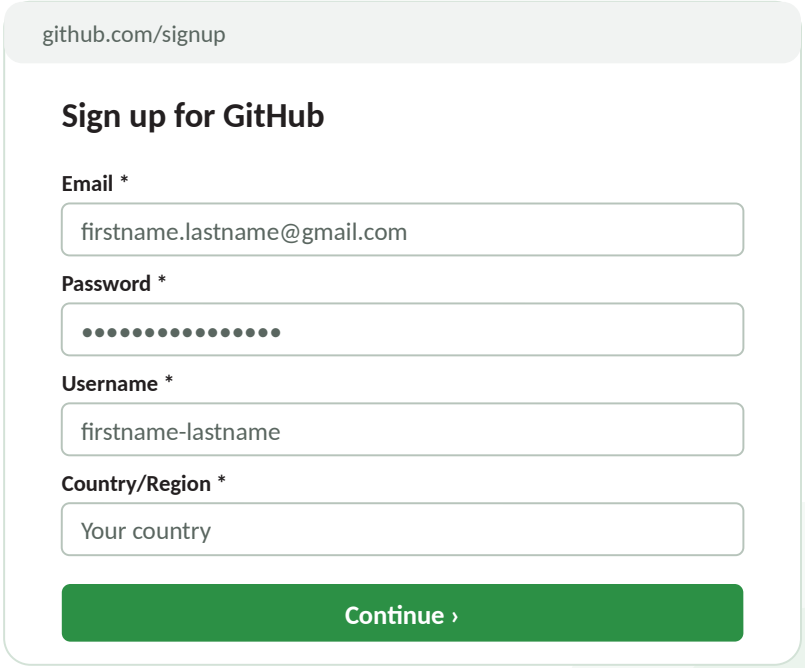
1. **Go to github.com/signup**
   Or click Sign up on github.com.

2. **Email**
   Your Gmail address, e.g. firstname.lastname@gmail.com.

3. **Password**
   At least 15 characters, or at least 8 with a number and a lowercase letter.

4. **Username**
   Letters, numbers and single hyphens. It appears in your URLs: keep it professional.

5. **Country/Region**
   Select it, then click Continue.



Schematic illustration — the live page may look slightly different.

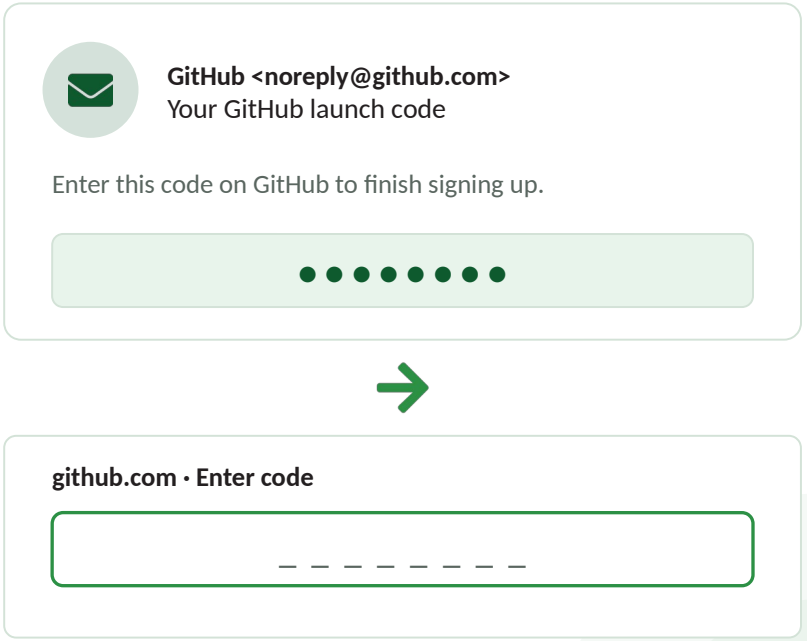
Prove you are human, then verify your email

6 **Solve the visual puzzle**
A short check, e.g. pick or rotate an image. Then click Create account.

7 **Open your Gmail inbox**
Look for the email from GitHub with your launch code. Check Spam or Promotions.

8 **Enter the code on GitHub**
Type the code in the GitHub page. If it expires, ask for a new one.

9 **Personalise or skip**
Answer the short questions or skip them: you reach your dashboard.



Schematic illustration — the live page may look slightly different.

Your account is ready: secure it

Keep your email verified

A verified email is needed for basic tasks such as creating a repository.

Turn on two-factor authentication

Strongly recommended by GitHub: Settings › Password and authentication.

Use a professional profile

Add your name and institution so teammates can find you.

The Free plan is enough

Public and private repositories at no cost: no payment details needed.

Sources: GitHub Docs — "Creating an account on GitHub", "Creating a strong password", "Verifying your email address".

Next: generate your API key ›

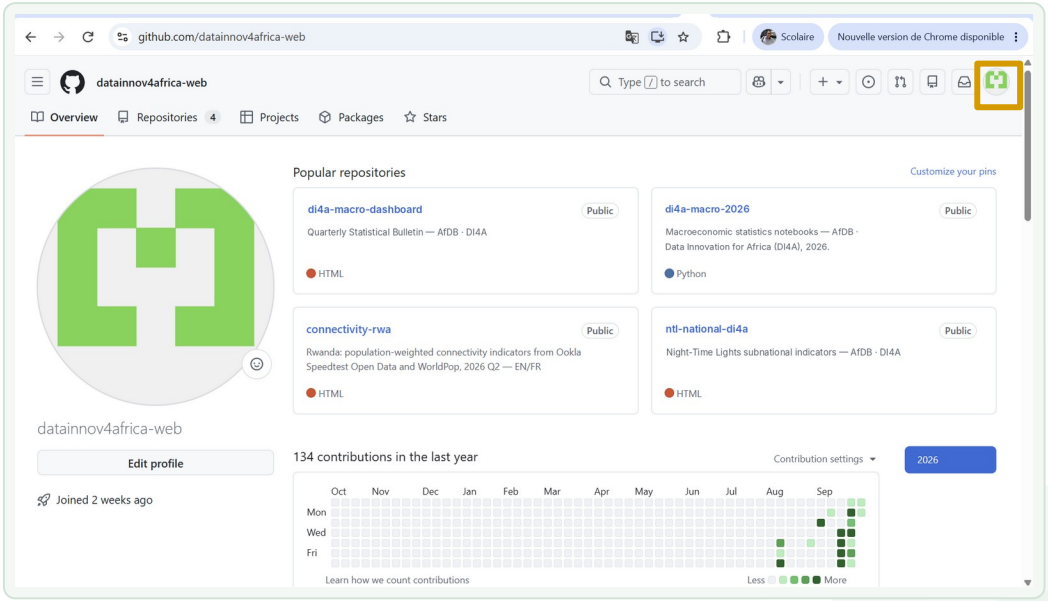
Open your account menu

Sign in to GitHub

Go to github.com and sign in with your GitHub account.

Click your avatar

Top-right corner of any GitHub page. It opens the account menu.



Screenshot: github.com — your profile page

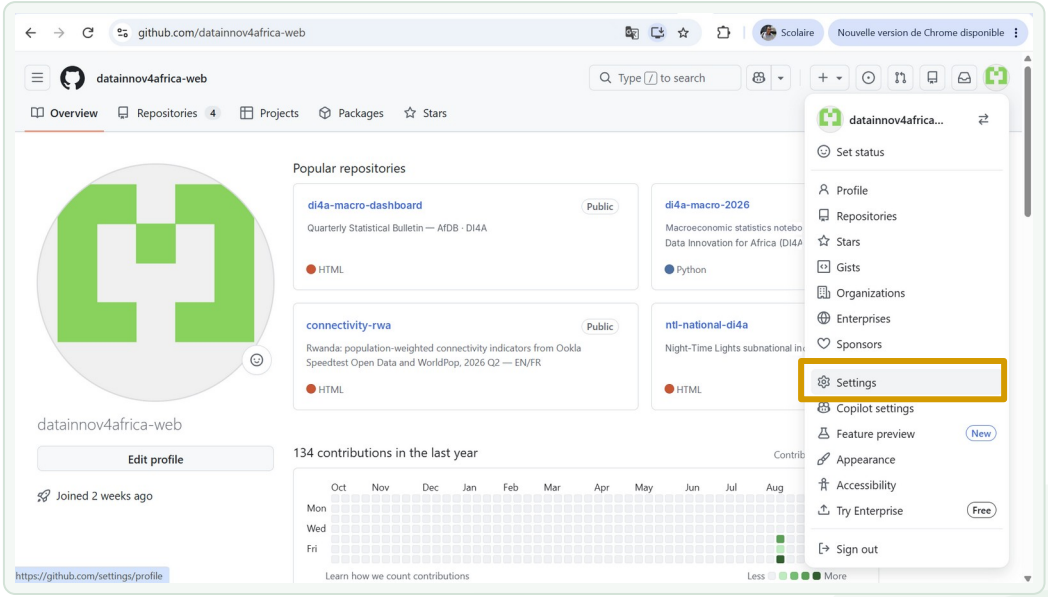
Go to Settings

Scan the menu

It lists Profile, Repositories, Stars, Organizations and more.

Choose Settings

It opens your profile settings at github.com/settings/profile.



Screenshot: github.com — account menu

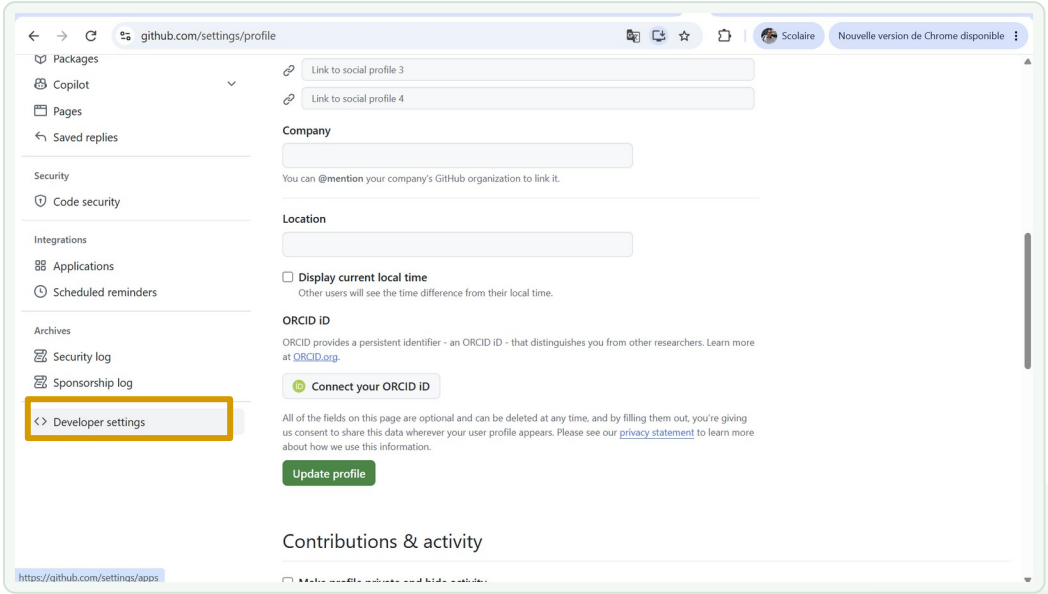
Open Developer settings

Scroll the left sidebar

Go all the way down, below the Archives section.

Click Developer settings

This is where GitHub Apps, OAuth Apps and access tokens live.



Screenshot: github.com/settings/profile

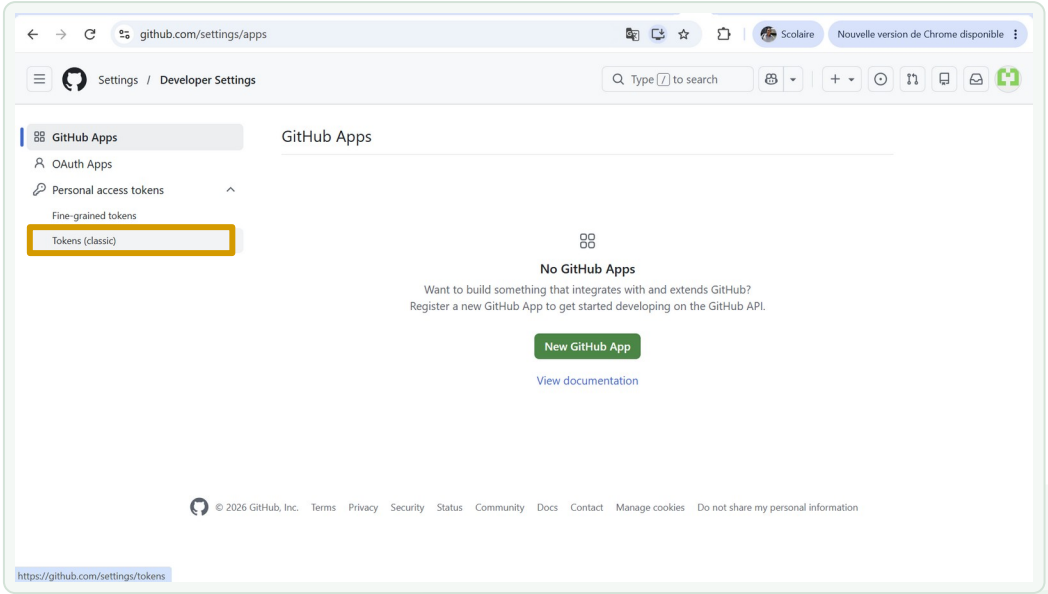
Select Tokens (classic)

Expand Personal access tokens

Two options appear: Fine-grained tokens and Tokens (classic).

Click Tokens (classic)

The token type used in this guide.



Screenshot: github.com/settings/apps

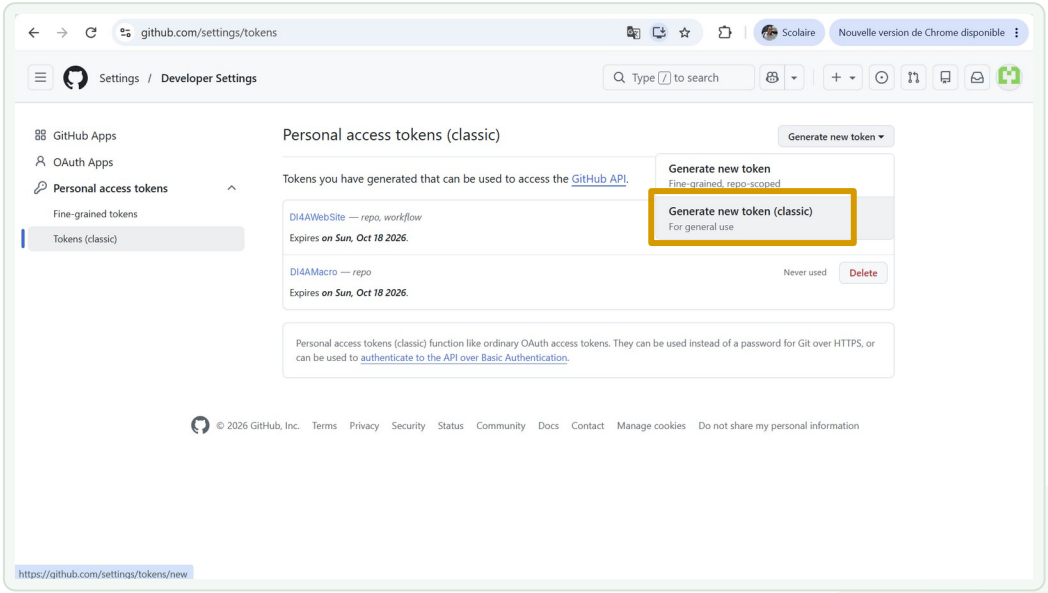
Generate a new classic token

Open "Generate new token"

The dropdown button above your list of tokens.

Pick "Generate new token (classic)"

Labelled "For general use". GitHub may ask you to confirm your password.



Screenshot: github.com/settings/tokens

Name it, set expiry, pick scopes

Add a note

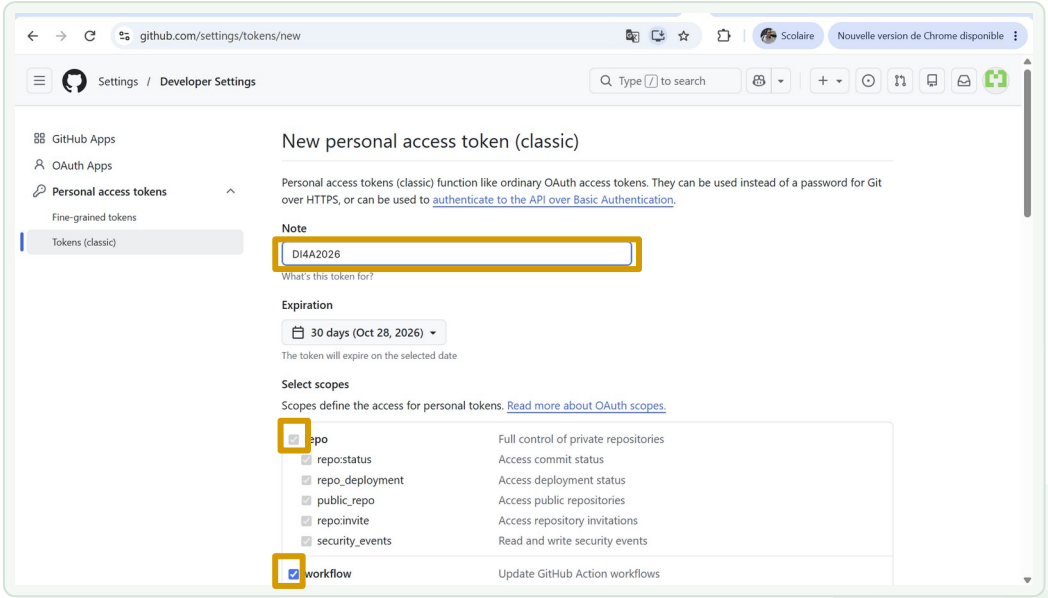
A name you will recognise, e.g. DI4A2026.

Set an expiration

30 days covers the hands-on work and follow-up.

Tick repo and workflow

Then scroll down and click "Generate token".



Screenshot: github.com/settings/tokens/new

Copy your token now

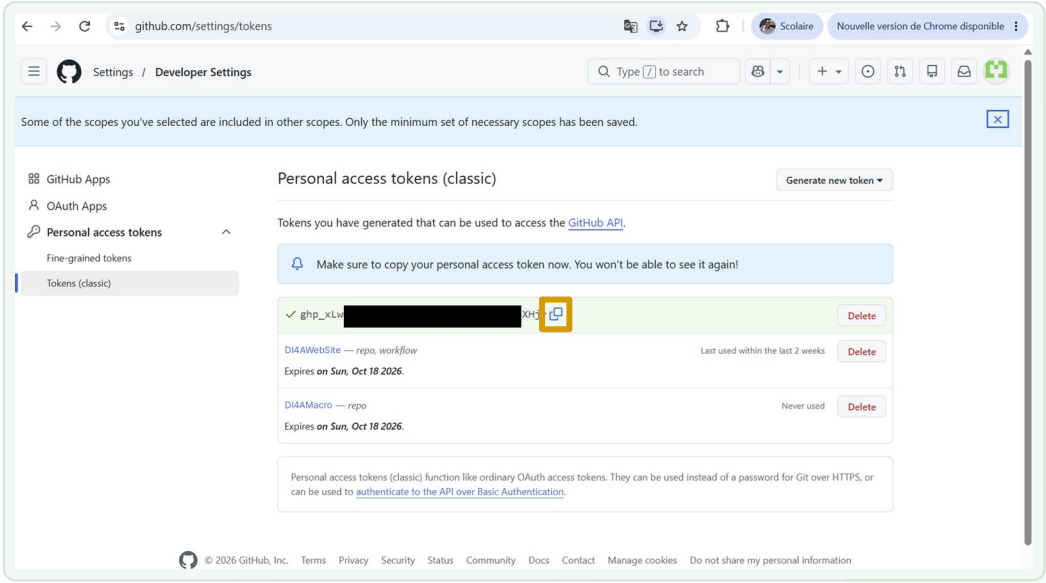
Click the copy icon

Next to the new token, which starts with ghp_.

Store it safely

In a password manager or an environment variable.

You won't be able to see it again. If you lose it, generate a new one.



Screenshot: github.com/settings/tokens — token masked

Keep your token safe

Never commit it

Keep tokens out of code and repositories. Use an environment variable or a GitHub Actions secret.

Always set an expiry

Short-lived tokens limit the damage if one leaks. Regenerate when it expires.

Grant minimum scopes

Tick only the scopes your script needs. For this guide: repo and workflow.

Revoke if exposed

Developer settings › Tokens (classic) › Delete, then generate a new token.

Source: GitHub Docs, "Managing your personal access tokens".

Sources

GitHub Docs — Managing your personal access tokens

docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens

Screenshots: github.com, captured with the datainnov4africa-web demo account (token value masked).

YOU ARE READY

Your token unlocks the GitHub API



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